MAR ATHANASIUS COLLEGE OF ENGINEERING
KOTHAMANGALAM

(Government Aided Autonomous Institution)

CERTIFICATE

MEDICAL SPECIALTY CLASSIFICATION USING TF-IDF AND ENSEMBLE MACHINE LEARNING

Certified that this is the bonafide record of the Mini Project work carried out by

JIPHIN GEORGE

Reg No: MAC25MCA-2033

during the third semester, in partial fulfilment of the requirements for the award of the degree of

Master of Computer Applications

of

APJ Abdul Kalam Technological University, Thiruvananthapuram

Prof. Biju Skaria
Faculty Guide

Dr. Sonia Abraham
Head of the Department

Dr. Manu John
Project Coordinator

Internal Examiner

Acknowledgement

I express my deepest praise, gratitude, and devotion to God Almighty for His boundless grace, wisdom, and blessings that guided me throughout the conceptualization, execution, and successful completion of this mini project.

I place on record my sincere thanks and respectful gratitude to Dr. Sonia Abraham, Professor and Head of the Department of Computer Applications, for her constant support, valuable encouragement, academic counsel, and for ensuring access to all necessary computing and software resources.

I am immensely indebted to my esteemed faculty guide, Prof. Biju Skaria, Associate Professor in the Department of Computer Applications, whose insightful guidance, rigorous technical feedback, continuous encouragement, and detailed supervision were vital to every stage of this research. His constructive suggestions during data preparation, feature leakage auditing, and model benchmarking contributed immeasurably to the success of this study.

I also extend my heartfelt thanks to the Project Coordinators, Dr. Manu John and Prof. Minu Maria Joseph, and all the faculty and non-teaching members of the Department of Computer Applications for their timely support, co-operation, and assistance throughout the semester.

Finally, I convey my deepest appreciation to my beloved parents, family members, and friends for their enduring patience, unconditional love, moral support, and motivation, which served as a constant pillar of strength throughout this project.

Jiphin George